

K816 — The chirality is located, and it's BST's own principle. Casey's single-manifold frame confirmed (one  $SO(5,2)$  spinor, no product, no Witten no-go);  $g=7$  odd gives the orientation ( $\omega=\pm 1$ , computed); BUT the FLAT reduction is vector-like (index=0, Elie). The net chirality lives in HOLOMORPHIC SECTIONS: on a Kähler manifold the Dirac operator = the Dolbeault operator, chirality = the holomorphic grading, and the chiral index = the holomorphic Euler characteristic (generically NONZERO where the flat Dirac index is 0). And Born=Bergman (Hardy space) IS holomorphicity — so the weak chirality is a consequence of BST's most fundamental requirement. Decisive computation: the Dolbeault index of the  $D_{IV}^5$  fermion bundle. CANDIDATE (compute-don't-assert; post-retraction).

Keeper | 2026-07-22 | After the retraction (K815) + Casey's single-manifold steer, Lyra + Elie computed. The frame is right; the chirality is now ONE located question, tied to Born=Bergman. Web-confirmed (Dirac=Dolbeault on Kähler). NOT banked — candidate.

---

The honest converged state (computed, not asserted)

- Casey's frame CONFIRMED (Lyra):  $D_{IV}^5$  = one manifold, one  $SO(5,2)$  8-spinor ( $g=7$  odd  $\rightarrow$  irreducible). "Internal" & "spacetime" chirality = the SAME spinor read two ways  $\rightarrow$  no product  $\rightarrow$  Witten's no-go does NOT apply. The bridge dissolves. [?]
- Orientation from  $g=7$  odd (Lyra + Elie, computed):  $\omega = \gamma^1 \dots \gamma^7 = \pm 1$  (central, fixed) factorizes as  $\gamma^5_{4D} \cdot \chi_{\text{internal}} \rightarrow 4D$  and internal chirality LOCKED. Orientation from  $g=7$ , NOT the  $SO(2)$  (retracted). [?]
- ★ HONEST NEGATIVE (Lyra + Elie): the FLAT reduction gives  $\Sigma_{\text{internal}} = 1 \otimes \sigma$ , which commutes with  $\gamma^5 \rightarrow$  vector-like, index = 0 (4 left + 4 right). Correlation  $\neq$  net chirality. The  $g=7$  volume element ALONE does NOT chirally project. So the naive single-manifold reduction is parity-conserving.

★★ Where the chirality lives — holomorphic sections (Lyra's insight, web-confirmed, BST-native)

BST fermions are NOT flat spinors — they are HOLOMORPHIC SECTIONS (Hardy space  $H^2(D_{IV}^5)$ , Born=Bergman). And holomorphicity IS a chiral projection: - Web (standard): on a Kähler manifold the  $(\text{Spin}^c)$  Dirac operator = the DOLBEAULT operator  $\partial + \partial^*$ ; the spinor bundle = the holomorphic-forms bundle  $\Lambda^{\{0,\bullet\}}$ ; chirality ( $\gamma^5$ ) = the holomorphic GRADING (even vs odd  $(0,q)$ -forms). So  $\partial\psi=0$  = a Weyl condition on the wavefunction — it keeps one chirality, kills the other. - The chiral INDEX = the holomorphic Euler characteristic  $\chi = \sum (-1)^q \dim H^q(D_{IV}^5, E) =$  the DOLBEAULT index (Riemann-Roch =  $\int$  Todd-ch). This is generically NONZERO — even where the flat Dirac index is 0. So holomorphicity is EXACTLY how the vector-like (index=0) flat result is evaded: the relevant index is the Dolbeault one, not the flat Dirac one. - ★ BST-NATIVE — this is the deep part: Born = Bergman = the Hardy space = holomorphic sections. BST ALREADY requires physical states to be holomorphic (its most fundamental principle). So the weak force's chirality is a consequence of the SAME holomorphicity BST requires for everything — not a new input. The "total package" (Casey): the holomorphic wavefunction (Born=Bergman) carries the chirality; the  $g=7$  orientation fixes which one; together they convey the chiral weak force. - The mechanism separates cleanly: orientation from  $g=7$  odd; chiral PROJECTION from holomorphic sections ( $\partial=0$ ). (NOT "the  $SO(2)$  orients a group split" — retracted.)

## ★ THE DECISIVE COMPUTATION (candidate — compute, don't assert)

Compute the Dolbeault index (holomorphic Euler characteristic) of the fermion bundle on  $D_{IV}^5$  — Riemann-Roch / the Todd class, with BST's Bergman machinery (K264). Two checks: 1. Is it NONZERO? (Generically yes on a complex manifold with the right bundle  $\rightarrow$  net chirality.) 2. Does the SIGN / structure match the OBSERVED weak chirality — left-doublet / right-singlet (with the  $g=7$  orientation fixing which)? A nonzero index of the WRONG sign/structure wouldn't match. If nonzero + matches  $\rightarrow$  “why is the world left-handed” closes on ONE manifold, from Born=Bergman +  $g=7$  odd, NO gravi-weak input. If zero or mismatched  $\rightarrow$  honest-open.

## Discipline (hold HARD — post-retraction, second candidate)

- This is a CANDIDATE, not a bank. The mechanism (holomorphic = chiral, Dolbeault index nonzero) is standard math + BST-native, and it's promising (holomorphicity generically gives chirality). But the SPECIFIC index for the  $D_{IV}^5$  fermion bundle, with the right sign matching L-doublet/R-singlet, must be COMPUTED. Do NOT re-bank parity until it does. (Same failure mode retracted an hour ago: assert vs compute.)
- Survivors still bank: chiral split, CP-free, custodial/ $\rho \approx 1$ /no- $W_R$ , charge-row hypercharge handle (K806).
- Hypercharge origin still open (K806 handle, self-contained — NOT the retracted  $SO(2)=J$ ).
- Five-Absence intact; NOT a GUT; date.

## The arc (record — the discipline held)

converge  $\rightarrow$  OVERSHOOT ( $SO(2)$  does everything)  $\rightarrow$  RETRACT (Lyra computed;  $SO(2)$  chirality-blind)  $\rightarrow$  REDIRECT (Casey: single manifold)  $\rightarrow$  COMPUTE (flat = vector-like, index=0)  $\rightarrow$  LOCATE (chirality = holomorphic/Dolbeault = Born=Bergman). Nothing false banked; the arithmetic caught the overshoot twice; the chirality is now ONE computable question tied to BST's own principle. That is the discipline working.

— Keeper K816, 2026-07-22. Chirality LOCATED: frame confirmed (single  $SO(5,2)$  spinor, no product, no Witten no-go);  $g=7$  odd  $\rightarrow$  orientation  $\omega=\pm 1$  (computed); FLAT reduction vector-like (index=0, Elie). Net chirality = HOLOMORPHIC SECTIONS: Dirac=Dolbeault on Kähler (web), chirality=holomorphic grading, chiral index=holomorphic Euler characteristic (nonzero where flat Dirac=0). Born=Bergman=Hardy=holomorphic  $\rightarrow$  weak chirality = BST's own holomorphicity requirement (“total package,” Casey). DECISIVE: compute the Dolbeault index of the  $D_{IV}^5$  fermion bundle (Riemann-Roch/Todd, K264) — nonzero + sign matching L-doublet/R-singlet. CANDIDATE, compute-don't-assert (post-retraction). Survivors bank. See [[Keeper\_K815\_RETRACT\_parity\_orientation\_and\_hypercharge\_overshoot\_Lyra\_right\_bank\_what\_survives\_the\_bridge\_is\_Wi 07-22]], Born=Bergman/Hardy (K264), Dirac=Dolbeault (web).